

From Independent Scientist to Novel Performer

A Small-Business Operating, Milestone, and Capitalization Plan for a Phase 1 LLM-Advised Robotic Whipple (ChemicalQDevice)

Draft 1.0

Repository v4.5.0

20 figures

12 milestones

SBIR Phase I

\$306,000

9 months, 5 milestones

SBIR Phase II

\$1,300,000

24 months, 7 milestones

Five-year programme

\$3,500,000

60 months, direct

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Independent research paper and practical adoption guide. It is not medical or regulatory advice and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. All figures derive from the author's repository sources and are illustrative unless tied to a cited reference.

Disclaimer: This work is independent and is not endorsed or sponsored by any trial sponsor, CRO, site, IRB, regulator, or medical society; and was adapted using Claude Code Opus 5.

The paper (v1.0) is at <https://doi.org/10.5281/zenodo.xxxxxxxx> and the repository (v4.5.0) is at <https://github.com/kevinkawchak/physical-ai-oncology-trials>. The paper files are deposited in funding/capitalization-plan.

Keywords Small business innovation research; Novel performers; Capitalization plan; Milestone schedule; Indirect cost; 21 CFR part 54; Pancreatic ductal adenocarcinoma; Robotic pancreaticoduodenectomy; Daraxonrasib; Physical AI

Table of Contents

Abstract	2
Executive Summary	2
How to Read This Document	2
1 The Novel-Performer Case	5
1.1 Three Clauses, One Applicant	5
1.2 Why the Pioneer Language Does Not Fit an Entity	5
1.3 Why the Novel-Performers Chapter Does Not Fit Either	5
1.4 The Clause That Fits, and What It Is Worth	6
2 Placeholder	8
3 Placeholder	9
4 Placeholder	10
5 Placeholder	11
6 Placeholder	12
7 Placeholder	13
8 Placeholder	14
9 Placeholder	15
10 Placeholder	16
11 Placeholder	17

Abstract

[DRAFTING INSTRUCTION] Write 200 to 240 words. Open with the conversion: the author has filed ten funding applications as an independent scientist, and one of the ten, application 05, is addressed to the only mechanism in the set built for a company. State the three numbers that govern the paper, \$306,000, \$1,300,000 and \$3,500,000, and the two derived ones, \$1,396,000 of direct work inside the award and \$2,104,000 outside it. Name the clause: not the report's Pioneer language and not its NOVEL PERFORMERS chapter, but the SBIR clause, which appears in Chapters I, III and IV. Close on what is being offered, which is a costed twelve-milestone schedule with an evidence artifact and a stop condition on every row. Source: `funding/pdac-funding-applications/final-apply/sections/sec-00-front.tex` for the abstract form, and `funding/science-golden-age/chunk-03-chapter-two-revitalizing-science-and-technology-enterprise.md` for the clause.

Executive Summary

[DRAFTING INSTRUCTION] Six short paragraphs, one per outline item A to E plus one on what the conversion costs. Each must carry at least one number. Do not restate the abstract. Sources: `funding/pdac-funding-applications/final-apply/sections/sec-08-budget-and-leverage.tex` for the budget frame, `funding/pdac-funding-applications/applications/app-05-nih-sbir-seed/` for the award split, and `funding/science-golden-age/chunk-08-annex-fiscal-year-2028-research-and-development-budget-priorities.md` for the 3 to 1 leverage target.

How to Read This Document

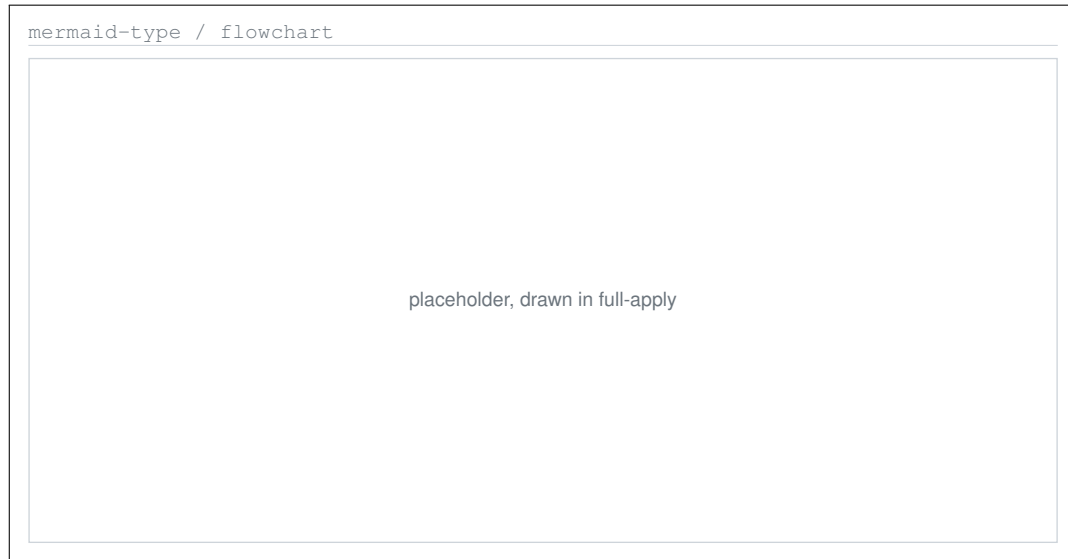
[DRAFTING INSTRUCTION] One short paragraph plus a four-row table mapping four readers to the sections each should read first: an SBIR program director, an ARPA-H programme manager, a private investor, and a UC San Diego intake committee. Source: `funding/pdac-funding-applications/final-apply/sections/sec-02-ten-applications.tex` for the reader-mapping form.

Fig	§	Platform	Source directory	The question it answers
1	1	Mermaid	mermaid/	Which clause does a company with no indirect-cost base qualify under?
2	1	D2	d2/	What kind of institution is a 2.6 FTE firm, on the report's own axes?
3	1	Graphviz	graphviz/	What does the same direct work cost under three overhead regimes?
4	2	Diagrams	diagrams-pyhton/	What is the shape of what the company holds today?
5	2	D2	d2/	What is owned, licensed, contracted, and absent, with terms?
6	2	PlantUML	plantuml/	What may the sponsor do alone, and what needs an institution?
7	3	Mermaid	mermaid/	What must hold at month nine for Phase II to begin?
8	3	D2	d2/	What does the same programme cost under two mechanisms?
9	3	Graphviz	graphviz/	Which work packages does the money reach, and which not?
10	4	PlantUML	plantuml/	Which capital positions may be held while a participant is on study?
11	4	D2	d2/	Where does every dollar come from, and what separates the tiers?
12	4	Mermaid	mermaid/	In what order is a private round executed during enrollment?
13	5	Mermaid	mermaid/	When does each milestone open, close, and produce its artifact?
14	5	Graphviz	graphviz/	What has to fail, and in what combination, to stop the programme?
15	5	PlantUML	plantuml/	What do three parties do at once to close one milestone?
16	6	D2	d2/	Which published quantities can a funder check before committing?
17	6	Graphviz	graphviz/	Where does each published number land in a filing?
18	7	Diagrams	diagrams-pyhton/	Who is employed, who is contracted, who is contributed?
19	8	Mermaid	mermaid/	What did the July and August 2026 contacts actually unlock?
20	10	Diagrams	diagrams-pyhton/	What survives if the programme stops, and who holds it?

The twenty figures, their section, their platform, the directory that carries each specification, and the one question each answers. Five mermaid, three plantuml, five d2, three diagrams, four graphviz, by purpose not quota.

Tab	§	Contents	The question it answers
1	0	Figure index, twenty rows	Where is each figure and what does it answer?
2	0	Table index, nineteen rows	Where is each table and what does it answer?
3	0	Four readers, four entry points	Which section should this reader open first?
4	1	Three clauses against four eligibility tests	Which clause survives the filter, and why?
5	1	Three overhead regimes on one direct base	What is the premium for routing through an institution?
6	2	The owned register, thirteen rows with DOIs	What exists today and where is it deposited?
7	2	The absent register, seven rows with signatories	What is missing and who alone can supply it?
8	3	The four-layer budget under two prices	What does the SBIR route reach, and what does it not?
9	3	Phase I line items, eight rows to \$306,000	What does nine months of Phase I actually buy?
10	3	Phase II line items, ten rows to \$1,300,000	What does twenty-four months of Phase II buy?
11	4	The §54.2 and §54.4 disclosure triggers	What must be disclosed, at what threshold, on which form?
12	4	The capitalization table across five stages	Who owns what, and does the SBIR test still pass?
13	5	The twelve milestones, costed and gated	What is the evidence, the cost, and the stop condition?
14	5	Milestone artifacts against funder mechanism	Which mechanism pays for which milestone?
15	6	Six checkable quantities with limitations	What can a reviewer verify against a published source?
16	6	Cost and schedule against industry benchmarks	What did the prior work cost, and what does it cost elsewhere?
17	7	The 2.6 FTE by employment status	Who is on the payroll and who is not?
18	9	Risks, likelihood, and the mitigation held	What could go wrong and what is already in place?
19	11	Abbreviations, two column pairs	What does this term mean in this paper?

The nineteen tables, their section, their contents, and the one question each answers. Every table is set to the full body measure with ragged-right fixed columns, so no cell shows a stretched interword gap.



Three clauses of one report, four eligibility tests, and the only clause a San Diego firm with no indirect-cost base survives. The two that fail, fail on size, not on merit or on subject matter.

1 The Novel-Performer Case

1.1 Three Clauses, One Applicant

[DRAFTING INSTRUCTION] *Open by naming the three candidate clauses and quoting each in one sentence. Clause 1 is the individual-scientist language of Chapter I, which the ten prior applications were written under. Clause 2 is the NOVEL PERFORMERS section of Chapter II. Clause 3 is the SBIR clause, which appears three times: in the Chapter I summary of recommendations, in Chapter III on deploying SBIR and STTR strategically to couple federally-seeded companies with the scientific enterprise, and in Chapter IV on SBIR opening doors for technician-founded ventures. Quote each from `funding/science-golden-age/chunk-01-front-matter-and-summary.md`, `funding/science-golden-age/chunk-03-chapter-two-revitalizing-science-and-technology-enterprise.md`, `funding/science-golden-age/chunk-04-chapter-three-securing-dominance-in-critical-and-emerging-technologies.md` and `funding/science-golden-age/chunk-05-chapter-four-science-and-technology-better-lives-of-all-americans.md`. Cite `kratsios2026sciencegoldenage`.*

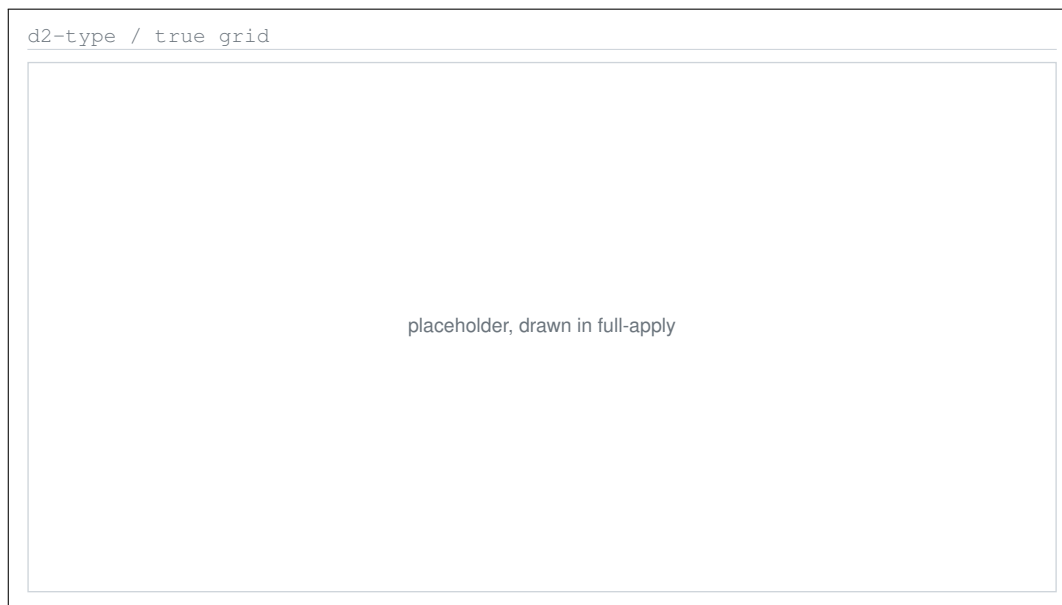
[DRAFTING INSTRUCTION] *After Figure 1, set Table 4: the three clauses against the four eligibility tests T1 to T4, with the ChemicalQDevice answer in the last column. The four tests and their report basis are given in `funding/capitalization-plan/mermaid/fig-01-clause-eligibility-filter.md`. Use five columns: test, question, report basis, this company, verdict.*

1.2 Why the Pioneer Language Does Not Fit an Entity

[DRAFTING INSTRUCTION] *Two paragraphs. The first states what changed: the applicant is now a California corporation and not a person, so a person-based mechanism has no subject to fund. The second is the honest concession, which must not be softened: the individual-scientist framing is what made the ten prior applications distinctive and is what the report actually rewards. Name the cost and forward-reference §9. Source: `funding/pdac-funding-applications/final-apply/sections/sec-01-golden-age-mandate.tex`.*

1.3 Why the Novel-Performers Chapter Does Not Fit Either

[DRAFTING INSTRUCTION] *One paragraph with three numbers taken from the chapter itself: tens of millions of dollars, coordinated teams of ten to a hundred people, and timelines of half a decade. Set them beside this company's \$1,606,000, 2.6 FTE and 33 months. The conclusion is that the chapter describes a real gap that this company does*



The report's own seven activities, its four institutional forms, and a fifth column it does not carry. An SBIR firm of 2.6 people is well suited to two of the seven and unsuited to two others.

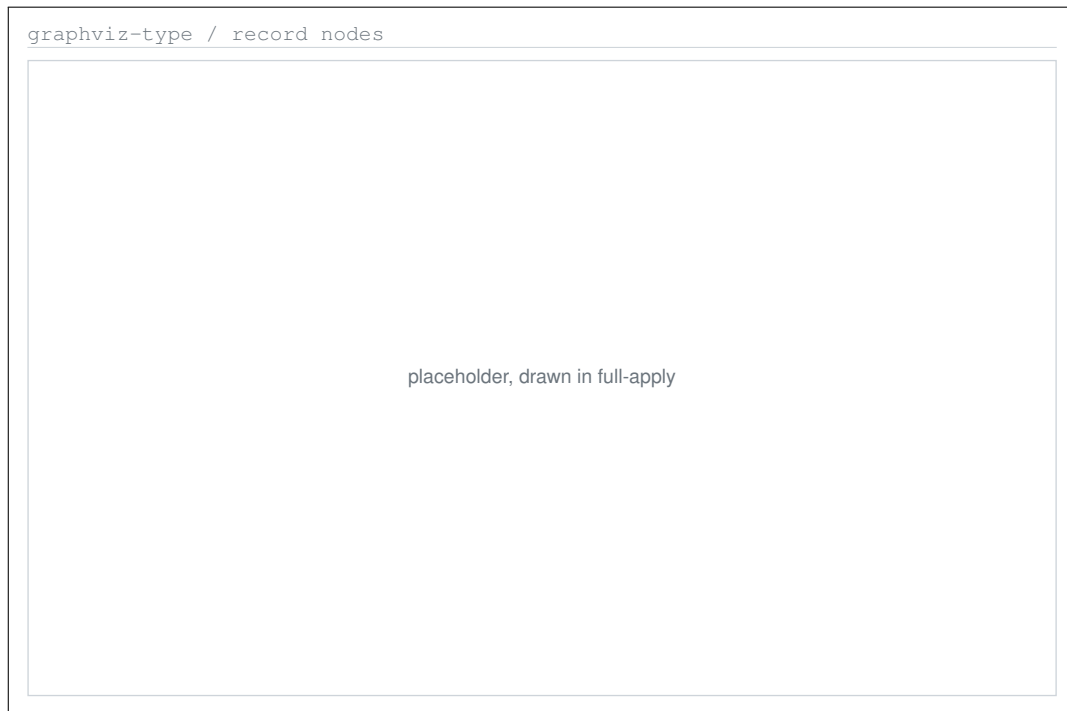
not fill, and that claiming otherwise would be the kind of overreach the report is written against. Source: funding/science-golden-age/chunk-03-chapter-two-revitalizing-science-and-technology-entprise.md, the NOVEL PERFORMERS section.

[DRAFTING INSTRUCTION] After Figure 2, one paragraph reading the fifth column: two well suited, three partly suited, two less suited, and the observation that a firm claiming to be well suited to all seven would be claiming to be a university, a corporate lab, a federal lab and an FRO at once. The row-by-row reasoning is in funding/capitalization-plan/d2/fig-02-institutional-form-grid.d2.md.

1.4 The Clause That Fits, and What It Is Worth

[DRAFTING INSTRUCTION] Open with the structural fact: this company has no negotiated indirect-cost rate agreement, no facilities to recover, and no administrative layer to fund. Then price it. Figure 3 and Table 5 carry the arithmetic from funding/capitalization-plan/graphviz/fig-03-indirect-cost-decomposition.gv.md. Cite nihsbirseed for the 40 percent allowance and ecfr2026uniformguidance for the de minimis rate.

[DRAFTING INSTRUCTION] Set Table 5 immediately after Figure 3: seven rows, four columns (line, route A university, route B full allowance, route C this plan), ending in the premium row. Every figure must reconcile; the checks are listed in funding/capitalization-plan/graphviz/README.md. Close the section with two sentences on what the 33.1 percent premium means to a program officer who has a fixed appropriation.



The same 1,396,000 of direct work priced three ways. A university F and A rate at 57 percent adds 741,000, the full SBIR allowance adds 695,000, and the rate this plan claims adds 210,000 in all.

2 Placeholder

3 Placeholder

4 Placeholder

5 Placeholder

6 Placeholder

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9 Placeholder

10 Placeholder

11 Placeholder
